ODUGUWA DAMILOLA JOHN

oduguwadamilola40@gmail.com ◇ linkedin.com/in/oduguwa-damilola ◇ damilojohn.github.io ◇ github.com/damilojohn

ML/AI Engineer Machine Learning Engineer specializing in LLM systems, ML infrastructure, and production backend engineering. Experience building distributed training systems, LLM-powered applications, and scalable cloud services.

EXPERIENCE

Lead Software Engineer March 2026 – Present
PocketCoach *Dallas, USA (Remote)*

- Own the backend and AI platform powering an AI tutoring product used by 20 schools and 2,500 + daily active students, with responsibility spanning application architecture, AI services, infrastructure, and deployment.
- Designed an event-driven question-bank generation system triggered by curriculum changes, combining structured LLM outputs, curriculum-aware RAG, and LLM-as-a-judge quality control to generate topic-level question banks powering practice, homework, and gamified learning features.
- Designed a fault-tolerant LLM gateway with semantic routers, transient-error retries, cross-model fallback on terminal failures, and token cost observability as the shared execution layer for all long-running AI workloads.
- Stood up a Redis/ARQ async worker platform decoupling email and document processing from the API which improved API response times by 40%, achieved near-exactly-once email delivery via dedup keys and backed retries, and let API and worker services scale independently.
- Engineered Paystack billing and subscription infrastructure with asynchronous webhook handling, idempotency guards against double-charges, reconciliation, automated renewals, dunning, grace periods, and double-entry ledger bookkeeping powering a parent-to-student performance reward system.
- Engineered multi-tenant RBAC (6 role tiers, JWT + Redis-backed refresh-token rotation) securing access across all 20 schools, maintained a parameterized GitHub Actions CI/CD pipeline with code quality tests via pre-commit hooks and ruff, automated testing with pytorch and multi-stage Docker builds, and container orchestration.

Machine Learning Engineer Oct 2025 – June 2026
Bluechip Technologies *Lagos, Nigeria*

- Developed LLM-powered schedule insights feature using agentic workflows and structured outputs, delivering AI-driven operational recommendations to scheduling officers at Nigeria's largest natural gas provider (NLNG).
- Designed a constraint optimization engine (Google OR-Tools CP-SAT) for cargo delivery scheduling, reducing schedule generation times by 60% and enabling rapid adaptation to operational changes.
- Built scalable event-driven backend using Azure Service Bus, concurrency primitives, and Redis with effective caching and cache invalidation strategies, improving API performance and reliability at scale.

Machine Learning Engineer (Contract) Jul – Oct 2025
Hobwise *Lagos, Nigeria*

- Built Airflow/Spark demand-forecasting pipelines and deployed model-serving infrastructure on GCP using Terraform and CI/CD across development, staging, and production.
- Containerized MLflow and FastAPI model-serving services; integrated with Kubeflow for distributed training orchestration and experiment tracking.

Software Engineering Intern Apr – Oct 2025
Stanbic IBTC (Standard Bank Group) *Lagos, Nigeria*

- Built a service-monitoring platform aggregating operational metrics across 60+ bank microservices, giving platform teams centralized visibility into service health and reliability.

- Contributed to a Face Verification service using Azure Cognitive Services for BVN-based identity verification across internal platforms.
- Built an encryption API mirroring the bank's production AES-GCM based encryption system, enabling the middleware team to securely test requests in local environments, and improving developer experiences.

Machine Learning Engineer Aug 2024 – Jan 2025
AIpplier *Louisiana, USA*

- Designed and shipped an LLM-based Computer Use agent (Playwright + LangGraph) to automate job applications across multiple job boards.
- Built a visual/DOM grounding system for LLM web navigation, combining bounding-box annotations with hierarchical DOM parsing to expose interactive page elements to the agent.

Machine Learning Engineer Aug 2023 – Feb 2024
Sienna AI *Lagos, Nigeria*

- Designed and deployed RAG systems (LangChain, LlamaIndex) and LLM agents for client-facing use cases, including blockchain (Solana) interaction, search, and database querying.
- Implemented custom function-calling for Llama2 and deployed multimodal LLM-serving REST APIs for client integrations.

Machine Learning Research Intern Oct 2022 – Apr 2023
Nithub Unilag AI Research *Lagos, Nigeria*

- Fine-tuned YOLOv5 for real-time traffic-lane detection as part of a traffic optimization system, demoed to stakeholders.
- Built a real-time CCTV-based prediction pipeline (AWS Greengrass, Kinesis, Lambda) for edge-triggered inference on live video streams.

SKILLS

- **Languages:** Python, Rust, C
- **AI/ML:** PyTorch, Slurm, FSDP, Triton, ONNX, LangGraph, RAG, MLflow, Apache Airflow, Apache Spark, OR-Tools.
- **Backend:** FastAPI, SQLAlchemy, PostgreSQL, Redis, MongoDB, WebSockets, ARQ
- **Infrastructure:** Docker, Kubernetes, Terraform, GitHub Actions, GCP, AWS, Azure, CI/CD

EDUCATION

BSc. Computer Engineering, University of Lagos *Expected August 2026*
Second Class Upper (Honours)

PROJECTS, RESEARCH & TALKS

SabiYarn — Pretrained a 125M-parameter LLM with a customized attention mechanism from scratch on 11 Nigerian languages via multitask learning; the model outperforms systems over 100× its parameter size on translation, NER, text diacritization, and sentiment analysis. Wrote the distributed multi-GPU training stack (FSDP across 2× A10G GPUs, ∼10.1B-token corpus) with custom Triton kernels, NUMA-aware OS optimizations. Developed training workflows and infrastructure for running evals on benchmarks and ablation tests of alternate model architectures (differential attention, multi-token prediction). *Published at ACL 2025.*

Zeno — Agentic search based book recommendation app built using FastAPI, AWS SQS, Exa, and LangGraph. Live demo.

Portal — Rust fork of `videocall.rs` that adds neural video compression, and a real-time audio-transcription service; built using the actix web framework, NATS, and fast-whisper. live.

Instructor at TRI AI Saturdays (Google DeepMind AI Research Foundation Course) Taught Week 3, "Training a Language Model": built a small LM from scratch (tokenizer through training loop) live for ∼600 participants; cohort of 3,134 across 83 countries.